

الإجابات على "كراس الإجابة"

Treat the following questions:

I- Choose the correct answer: (1pt/ correct answer)

1- Prolog is:

- a- An object oriented programming language
- b- A procedural programming language
- c- A logic programming language
- d- An http tag

3- An extent is which of the following?

- a- A keyword that indicates that the subclass inherits from a superclass
- b- A keyword that indicates that the superclass inherits from a subclass
- c- The set of all instances of a class within a database
- d- Only one instance of a class within a database

5- Which one of the following, is NOT a fragmentation type?

- a- Vertical fragmentation (VF)
- b- Horizontal fragmentation (HF)
- c- Mixed fragmentation (MF)
- d- Database cloud fragmentation(DCF)

7- Which of the following is not one of stage in the evolution of distributed DBMS?

- a- Unit of work
- b- Remote unit of work
- c- Distributed unit of work.
- d- Distributed request

9- Recursive rule is a rule whose the head predicate also appears in the body:

- a- True
- b- False

2- Which of the following RDBMS is qualified as object oriented?

- a- INFORMIX
- b- ORACLE
- c- O2

4- Which is an advantage of OODB?

- a- Extensibility
- b- Lack of experience
- c- Lack of standards
- d- Complexity

6- Which of the following is a disadvantage of replication?

- a- Reduce network traffic
- b- If the database fails at one site, a copy can be located at another site
- c- Each site must have the same storage capacity

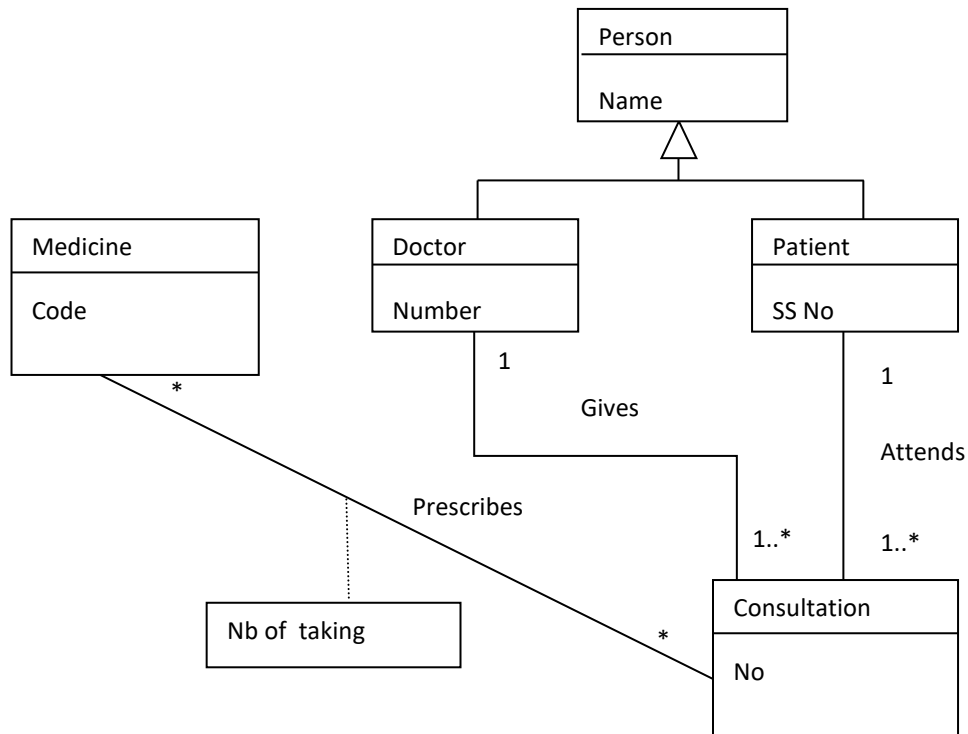
8- Deductive DBMS allows to derive the tuples of intentional predicates by use of rules:

- a- True
- b- False

10- Intentional predicate is a predicate whose instances are stored in the database as tuples:

- a- True
- b- False

II- Use ODL language to transform the following class diagram into an object database schema. (4 pts)



III- A bank has decided to distribute its database at the country level. The current centralized relational database schema is as follows: (6 pts)

Country	(<u>c</u> code, cname)
City	(<u>c</u> id, cname, ccode)
Customer	(<u>c</u> ustId, custName, cid)
Currency	(<u>c</u> ode, name)
Account	(<u>a</u> id, aname, balance, custId, code)
Employee	(<u>e</u> id, firstName, lastName, salary, cid)

- Countries are France, Spain, and Germany.
- The bank's head quarter is located at Germany. It contains all information about all the employees.

- Produce a distributed database design for this company.
- For each site, define each fragment in relational algebra and specify its type.
- Reconstruct the original global relations from their fragments.